

# CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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**Abstract.** University counseling spans relational curricula, cohort-specific fees, exemption policies, and narrative regulations, each requiring distinct evidence representations. Whether decomposing this heterogeneous task into specialized agents improves reliability over an equivalent single agent remains underexplored. We present CTU-Chat, a supervisor-routed multi-agent system assigning queries to four bounded specialists with scoped tools and representation-specific evidence paths: Neo4j graph traversal for curricula, structured lookup with deterministic calculation for financial data, and hybrid retrieval with neural reranking for regulations. Scenarios 1 and 2 use a 100-query held-out benchmark for architecture comparison and ablation, respectively. The former compares CTU-Chat with a matched single-agent baseline and a routed generic variant; the latter removes specialist policies, tool isolation, heterogeneous evidence, deterministic calculation, and route repair. Scenario 3 uses 50 separate cases to stress-test tool ambiguity across 11, 31, and 51 tools. In architecture comparison, CTU-Chat achieves a 10.0 pp higher Automated Fact Pass Rate ( $\text{Pass}_{\geq 50\%}$ ) point estimate (76.0% vs. 66.0%; paired 95% CI [+2.0, +19.0] pp; exact McNemar  $p=0.0414$ ) with 41.4% fewer recorded specialist input tokens, alongside an unlogged upstream routing overhead of ~650 tokens and sequential dispatch latency (10.65 s pipeline vs. 5.16 s post-retrieval baseline). Under registry expansion to 51 tools, the scoped specialist achieves 88.0% tool E2E Success (vs. 78.7% for Top-10 selection). These findings suggest that domain-scoped specialization and heterogeneous evidence allocation may improve fact-pass performance under the evaluated conditions, with measurable trade-offs in coordination latency and token consumption.

**Keywords:** Multi-Agent Systems · Supervisor Routing · Architectural Ablation · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

## 1 Introduction

At the opening of the 2025–2026 academic year, Can Tho University (CTU) reported 48,960 undergraduate students<sup>1</sup>. Its scale and evolving regulations create advisory needs spanning heterogeneous knowledge forms: (1) **academic curricula** encoding hierarchies and prerequisites; (2) **tuition schedules** varying by cohort and field; (3) **exemption and scholarship policies** combining eligibility conditions with rates; and (4) **administrative regulations** comprising lengthy narrative documents. Prior systems address individual domains—CAAS for IT admission counseling [8] and REBot for regulation advising [7]—but a single representation or retrieval method may be unsuitable across all domains.

RAG and GraphRAG augment LLMs with documents and knowledge graphs [6,3,2], while agentic architectures support tool use and multi-agent coordination [18,16,14]. These approaches do not prescribe how heterogeneous institutional evidence should be represented, routed, and processed. Known limitations in language-model arithmetic [12] further motivate deterministic calculation for financial queries.

*Central Hypothesis.* When a counseling system must handle diverse knowledge forms and semantically adjacent tools, routing queries to bounded specialists with specialized evidence paths can improve reliability relative to a single agent with equivalent capabilities, albeit with coordination latency. Whether this trade-off is favorable requires direct comparative evaluation.

These challenges highlight three primary research gaps: (1) limited evidence comparing multi-agent and single-agent architectures under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited evidence on how bounded specialization maintains reliability as the tool registry grows. We formulate three research questions:

- **RQ1:** How does the proposed multi-agent architecture compare with a functionally matched single-agent in end-to-end reliability, evidence grounding, and cost?
- **RQ2:** What performance changes are observed when specialist policies, tool isolation, representation-specific evidence access, deterministic computation, and route repair are individually removed or replaced?
- **RQ3:** How does bounded specialist routing affect tool-selection reliability as semantically overlapping tools expand the registry from 11 to 51 tools?

This work provides three primary contributions: (1) **C1**—a supervisor-routed bounded multi-agent architecture with four specialists, deterministic route repair, and partitioned tool privileges; (2) **C2**—an architecture-level comparative evaluation against a matched single-agent baseline and a routed generic variant; and (3) **C3**—architectural ablations of five components and an 11–51 tool ambiguity stress test.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the CTU-Chat architecture. Section 4 describes the experimental

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<sup>1</sup> CTU opening ceremony report, 5 September 2025: <https://en.ctu.edu.vn/news-t/can-tho-university-inaugurates-the-2025-2026-academic-year-embarking-on-a-new-era-and-affirming-its-role-as-the-leading-institution-in-education-training-science-and-technology-in-the-mekong-delta.html> (accessed 28 September 2026).

setup. Section 5 reports results. Section 6 discusses findings and limitations. Section 7 concludes.

## 2 Related Work

### 2.1 Vietnamese NLP and Retrieval Foundations

Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [9]; 85% of word types contain at least two syllables. While domain-adapted bi-encoders capture semantic paraphrasing, Vietnamese higher-education decrees rely on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes, which necessitates the coupling of dense embeddings with exact lexical matching.

BM25 [13] provides reliable exact token matching for rigid identifiers but cannot capture paraphrases. Dense retrieval [5,6] captures paraphrastic intent but may struggle with rare alphanumeric tokens. Reciprocal Rank Fusion (RRF) [1] merges BM25 and dense candidates without score calibration, while neural cross-encoders [10,17] jointly encode query–passage pairs for reranking. Adaptive-RAG [4] demonstrates complexity-based retrieval selection. These approaches support dynamic retrieval but do not allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

### 2.2 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples [11]. GraphRAG [2] leverages graph topology to uncover multi-hop associative paths that escape flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious relationships, motivating selective graph construction restricted to domains with well-defined relational schemas.

### 2.3 Agentic RAG and Multi-Agent Orchestration

ReAct [18] demonstrates interleaved reasoning and action, AutoGen [16] provides a multi-agent conversation framework, and Toolformer [14] studies tool invocation. For institutional counseling, these capabilities raise design risks without explicit constraints: broad tool exposure increases errors; peer-to-peer exchanges increase coordination cost; a shared index mixes evidence across domains. Controlled comparisons against functionally matched single-agent baselines remain limited, as most evaluations report absolute performance without isolating the contribution of multi-agent decomposition from underlying capabilities.

### 2.4 Educational Counseling Systems

Educational chatbots assist in advising and learning support [15]. CAAS supports admission counseling for IT programs [8], while REBot combines retrieval with category-guided GraphRAG for academic regulations [7]. CTU-Chat differs from prior systems

in three aspects: (1) *Task scope*: representing 111 distinct program codes from 113 curriculum files in relational property graphs, with deterministic tuition arithmetic; (2) *Multi-agent routing*: a LangGraph supervisor routes to four bounded specialists with isolated evidence spaces; and (3) *Architectural evaluation*: direct comparison against functionally matched baselines with component ablations.

## 2.5 Research Gaps

An analysis of the reviewed literature reveals three primary gaps: (1) limited architecture-level evidence comparing multi-agent and single-agent systems under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited robustness evidence under tool ambiguity as semantic neighbors accumulate. These gaps motivate the evaluation design in Sections 4–5.

## 3 Proposed Model

CTU-Chat combines three design principles: centralized coordination, heterogeneous knowledge allocation, and representation-aware evidence routing. A contract-checked supervisor assigns one specialist per query. Depending on the domain, the specialist uses graph traversal for curricula, structured lookup and calculation for fees, or retrieval for narrative policies.

### 3.1 Overall Architecture

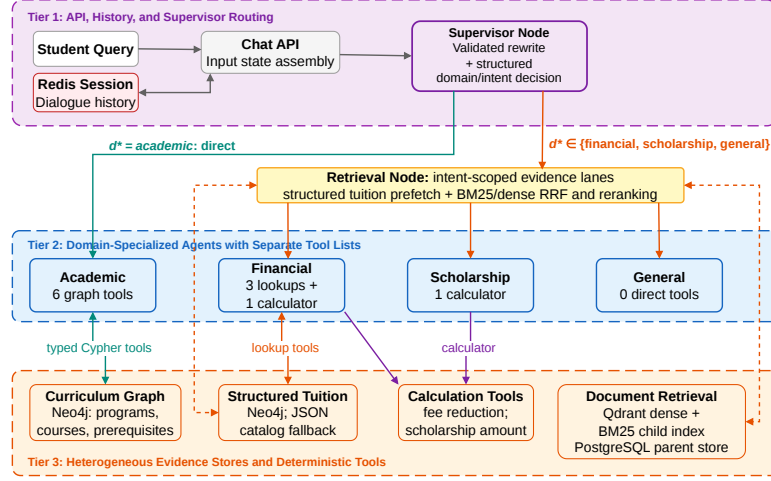
Figure 1 illustrates three coordinated tiers: (1) the Interaction and Centralized Supervisor Tier, (2) the Domain-Specialized Multi-Agent Core, and (3) the Heterogeneous Knowledge Storage and Tool Tier.

CTU-Chat uses a typed `AgentState` for rewrites, routes, evidence, and responses. An intent-owner contract checks supervisor output before conditional dispatch: academic queries enter the graph agent; other domains use intent-scoped retrieval. Specialists see only assigned tools and return to END without peer exchanges. The fixed-query benchmark in Section 4 bypasses Redis history and query rewriting used in live multi-turn serving.

### 3.2 Neo4j Knowledge Graph and Cypher Tools

The Neo4j graph is built from 113 curriculum Markdown files representing 111 distinct program codes. The resulting graph stores approximately 7,000 nodes across 10 labels: curriculum entities (e.g., `Program`, `Course`, `Block`, `PLO`, `PEO`, `Track`, `Job`) and tuition entities (`TuitionFee`, `TuitionPolicy`, `ExemptionRate`) connected by over 16,000 edges across 13 typed relationships encoding containment, prerequisites, program outcomes, tracks, occupations, tuition, and exemption bases.

The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies arguments. Each tool normalizes identifiers and calls a fixed, parameterized Cypher query implementing three query classes: *attribute lookup/search* (program codes, names,



**Fig. 1.** Implemented CTU-Chat architecture. Redis-backed history and validated query rewriting are used in live multi-turn serving; the offline benchmark uses fixed single-turn queries and bypasses these components. The supervisor’s structured decision is contract-checked before conditional dispatch. Academic queries enter the graph-enabled agent; other domains pass through intent-scoped retrieval.

comparisons), *structural traversal* (prerequisite chains up to ten hops, course intersection, reverse path search), and *structured financial lookup* (tuition, policies, exemption rates). This design binds agent actions to deterministic graph operations, reducing exposure to errors from unconstrained Cypher generation.

### 3.3 Routing and Deterministic Arithmetic

For a query  $q$  and its dialogue predecessor, the supervisor first attempts a validated standalone rewrite, then predicts domain  $d_{\text{raw}}$  and intent  $i_{\text{raw}}$  under a structured schema, falling back to **general/other** on parsing failure. The ownership contract resolves rule conflicts or incompatible domain–intent pairs. Academic queries proceed directly to the graph agent; other domains build intent-scoped retrieval lanes, with financial queries first attempting Neo4j tuition lookup and then JSON fallback.

**Deterministic Tuition Arithmetic.** For fee-reduction requests, the tool `tin_h_toan_hoc_phi` receives three parameters—per-credit tuition  $T$ , per-credit exemption basis  $B$ , and exemption percentage  $r \in [0, 100]$ —and calculates deductible credit amount  $D = B \times (r/100)$  alongside net payable tuition  $T_{\text{net}} = \max(0, T - D)$ . The tool clamps  $T_{\text{net}}$  at zero. The primary contribution is the integration of validated rewriting, constrained routing, and deterministic arithmetic into a unified multi-agent workflow.

### 3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

CTU-Chat stores curricula in Neo4j, accesses cohort-specific fees through structured lookup, and retrieves narrative regulations from documents. Financial requests try Neo4j tuition lookup with JSON fallback before the repaired intent selects metadata-filtered retrieval lanes.

For narrative regulations and scholarship criteria, BM25 and Qdrant search 400-character child chunks; parent sections (2,800-char target, 100-char overlap) are loaded from PostgreSQL and fused via RRF:  $s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} (60 + r_s(d))^{-1}$ . The full stack pools intent-governed and unfiltered candidates, deduplicates, and reranks with `bge-reranker-v2-m3`. Scores below 0.20 are removed unless fewer than two candidates remain; graph evidence is added under fixed quotas and up to seven passages are packed for generation.

### 3.5 Cross-Domain Query Handling

For cross-domain queries, CTU-Chat uses a single-owner, signal-based evidence-prefetch policy. The supervisor selects an owner by dominant intent; keyword signals add academic-program, academic-regulation, tuition, or scholarship retrieval lanes. The owner uses this evidence without calling another specialist. This bounds execution but may leave a needed tool or source outside the owner’s scope. The 20 cross-domain held-out queries test this policy.

## 4 Experimental Setup

### 4.1 Dataset and Annotation

We evaluate CTU-Chat on 200 administrative queries constructed through a document-grounded protocol over official CTU statutes. The 100-query development split supports index calibration and prompt engineering. The 100-query held-out set follows a predefined distribution: 40 direct, 20 within-domain multi-hop, 20 cross-domain, 10 entity-comparison, 5 temporal, and 5 adversarial queries (51 formal, 49 colloquial; 38 single-source, 62 multi-source). It contains 19 Academic, 28 Financial, 17 Scholarship, and 16 General single-domain queries; the remaining 20 combine evidence across domains. Candidate queries and ground-truth evidence were extracted from official statutes: one annotator established reference answers, required facts, and gold sources, while a second annotator independently verified all annotations; disagreements were adjudicated against official regulatory text. An audit found no cross-split duplicates (max 5-gram Jaccard = 0.1875). Each query is annotated along nine dimensions, including domain, evidence type, and tool requirements, to support fine-grained failure analysis.

### 4.2 Baselines and Configurations

**Baseline equivalence and execution bounds.** Scenario 1 configurations share Gemini 2.5 Flash Lite ( $T=0.0$ ), knowledge representations, deterministic calculator access, and the same retrieval output (BM25 + dense + RRF + `bge-reranker-v2-m3`,  $k=7$ ).

Shared retrieval gives identical Source Recall (78.6%) and Source AP (70.8%); tool outputs remain configuration-specific. Scenario 1 compares orchestration bundles, while Scenario 2 changes one component at a time. Multi-agent variants allow at most 3 tool calls, 2,048 output tokens, and 30-second timeouts. Reported tokens cover specialist execution; supervisor routing adds an estimated ~650 input tokens per query.

*Scenario 1 — Architecture Comparison (RQ1):* Three configurations are evaluated on the held-out set ( $N=100$  queries, 300 runs under greedy decoding  $T=0.0$ ): **S1-A:** Unified Single Agent with all 11 tools, no supervisor or routing; **S1-B:** Routed Generic Worker—supervisor routes to domain workers with scoped tools but a single generic prompt; **S1-C:** Full CTU-Chat with supervisor, contract repair, specialist prompts, scoped registries, and deterministic computation.

*Scenario 2 — Ablation (RQ2):* Five ablations each remove one component from Full CTU-Chat ( $N=100$  queries per ablation,  $T=0.0$ ): **S2-A1:** No Specialist Policy (generic worker prompt); **S2-A2:** No Tool Isolation (all 11 tools visible); **S2-A3:** Uniform Text-RAG Baseline (graph data serialized to text, traversal disabled); **S2-A4:** No Deterministic Calculation (LLM generates arithmetic directly); **S2-A5:** No Route Repair (raw supervisor output used directly).

*Scenario 3 — Tool Ambiguity (RQ3):* Fifty fixed cases are evaluated while 40 semantic-neighbor tools expand registries to 11, 31, and 51 tools. Four configurations are compared: Full-Registry Single Agent, Global Top-10 Single Agent, Routed Generic, and Routed Specialist. Each case uses 3 counterbalanced tool-order blocks, yielding  $50 \times 4 \times 3 \times 3 = 1,800$  decisions across counterbalanced blocks.

### 4.3 Evaluation Metrics

Reported metrics span four evaluation dimensions: (1) *Final-answer quality*: Primary evaluation combines lexical fact grounding with semantic LLM judging (Gemini 2.5 Flash Lite) and programmatic arithmetic checks. We measure continuous Answer Correctness ( $AC \in [0, 1]$ ), moderate Semantic Pass Rate ( $\text{Pass}_{.50}^{\text{Jdg}} = \mathbb{I}(AC \geq 0.50)$ ), strict Semantic Pass Rate ( $\text{Pass}_{.75}^{\text{Jdg}} = \mathbb{I}(AC \geq 0.75)$ ), and Answer Relevancy ( $AR \in [0, 1]$ ). For financial queries ( $N_{\text{fin}} = 50$ , comprising 28 single-domain tuition and 22 cross-domain fee queries), programmatic Numeric Exact Match (Num-EM) strictly evaluates final payable tuition: numbers are regex-extracted, normalized to integer VND values, and verified against ground-truth fee schedules to prevent false-positive lexical passes on flawed arithmetic. These are complemented by lexical Required-Fact Coverage (FactCov) and Automated Fact Pass Rate ( $\text{Pass}_{.50}^{\text{Lex}}$ ). (2) *Evidence retrieval*: Source Recall (Src.R) and Source AP (Src.AP) assess Scenario 1. Table 4 evaluates Hit@ $k$ , P@5, R@5, and MRR@10 over 100 queries after collapsing duplicate chunks. (3) *Execution reliability*: Tool Invocations per Query, Tool Selection Accuracy, deterministic Tool End-to-End Success (E2E), and route repair rate. (4) *Operational cost*: Specialist input/output tokens and wall-clock latency. Scenario 3 further tests robustness via deterministic *Tool E2E Success*.

#### 4.4 Statistical Protocol and Infrastructure

Evaluations use greedy decoding ( $T=0.0$ ) to minimize sampling variance. Paired differences across configurations are computed at the query level ( $N=100$ ); 95% CIs are estimated using 10,000 bootstrap resamples. Timing follows configuration boundaries: baseline S1-A latency measures agent invocation post-retrieval, while multi-agent variants measure pipeline latency inclusive of supervisor routing, route repair, and specialist execution. Reported tokens record specialist agent execution; upstream supervisor routing adds an estimated overhead of  $\sim 650$  input tokens per query.

Experiments run on Gemini 2.5 Flash Lite via Vertex AI ( $T=0.0$ ) with neural reranking on a GTX 1650. Neo4j 5.20 stores relational curricula; PostgreSQL stores parent chunks (2,800 characters, 100 overlap); Qdrant indexes child chunks (400 characters, 50 overlap) with a 768-dimensional vietnamese-bi-encoder. The pipeline uses bge-reranker-v2-m3, LangGraph StateGraph, and Redis 7. Latency excludes the first warm-up run. Prompt templates, tool contracts, and schemas are archived.

## 5 Results

### 5.1 Scenario 1 — Architecture-Level Comparison

Table 1 compares three arms on the 100-query held-out set. Shared pre-generation retrieval fixes Source Recall (78.6%) and Source AP (70.8%) across arms; answer and cost differences reflect their combined orchestration changes.

**Table 1.** Scenario 1: Architecture-level comparison ( $N = 100$ ).  $\text{Pass}_{.50}^{\text{Lex}}$  = Lexical Fact Pass Rate ( $\geq 50\%$  required facts).  $\text{Pass}_{.50}^{\text{Jdg}} / \text{Pass}_{\geq .75}^{\text{Jdg}}$  = LLM-Judge Semantic Pass Rates ( $\text{AC} \geq 0.50 / \geq 0.75$ ).  $\text{AC} / \text{AR}$  = Mean Answer Correctness / Relevancy. Num-EM = Numeric Exact Match on financial queries ( $N_{\text{fin}} = 50$ ). Fact = mean fact coverage (%). In/Out Tok = specialist agent tokens; supervisor routing ( $\sim 650$  tokens/query) is excluded. Bold = best reliability / lowest cost. Shared pre-generation retrieval fixes Source Recall (78.6%) and Source AP (70.8%) across arms.

| Arch.        | $\text{Pass}_{.50}^{\text{Lex}}$ | $\text{Pass}_{.50}^{\text{Jdg}}$ | $\text{Pass}_{\geq .75}^{\text{Jdg}}$ | AC          | AR          | Num-EM      | Fact        | In Tok      | Out Tok     | Lat. (ms)   |
|--------------|----------------------------------|----------------------------------|---------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Single Agent | 66.0                             | <b>61.0</b>                      | 33.0                                  | <b>58.6</b> | <b>80.9</b> | 60.0        | 57.5        | 6296        | 1530        | <b>5159</b> |
| Routed Gen.  | 66.0                             | 52.0                             | 29.0                                  | 48.1        | 66.7        | 54.0        | 54.2        | <b>3473</b> | <b>1415</b> | 8921        |
| CTU-Chat     | <b>76.0</b>                      | 59.0                             | <b>37.0</b>                           | 57.7        | 77.8        | <b>62.5</b> | <b>61.0</b> | 3690        | 2092        | 10652       |

**Pairwise bootstrap analysis** (10,000 resamples): (1) CTU-Chat vs. Single Agent: CTU-Chat achieves a 10.0 pp higher  $\text{Pass}_{\geq 50\%}^{\text{Lex}}$  estimate (76.0% vs. 66.0%, paired 95% bootstrap CI [+2.0, +19.0] pp; McNemar  $p=0.0414$ ), higher Fact Coverage ( $\Delta = +3.58$  pp, CI [−1.2, +8.5] pp), and 41.4% fewer specialist input tokens (differing boundaries: 10,652 ms end-to-end vs. 5,159 ms post-retrieval). (2) CTU-Chat vs. Routed Generic: CTU-Chat gains +6.85 pp Fact Coverage (CI [+0.7, +13.0] pp,  $p < 0.05$ ). Generic and Single Agent both pass 66.0% (Fact Coverage 54.2% vs. 57.5%). (3) Cross-domain queries: CTU-Chat passes 80.0% (vs. 70.0% Single, 65.0% Generic). Evaluating threshold sensitivity reveals trade-offs: CTU-Chat’s margin at 50% narrows at 60% (54.0% vs.

50.0%) and reverses at 75% (27.0% vs. 30.0%,  $p=0.5488$ , 95% CI  $[-10.0, +3.0]$  pp), showing neither architecture dominates at stringent lexical thresholds.

**LLM-as-a-Judge and Numeric Exact Match:** To complement lexical metrics, an independent LLM judge (Gemini 2.5 Flash Lite) and programmatic Numeric Exact Match (Num-EM,  $N_{\text{fin}}=50$ ) evaluated all 100 held-out queries. Continuous evaluation shows comparable Mean Answer Correctness across architectures: Single Agent achieves 58.6% versus 57.7% for CTU-Chat ( $\Delta = -0.90$  pp, 95% CI  $[-7.2, +5.3]$  pp) and 48.1% for Generic. On moderate semantic threshold ( $\text{Pass}_{.50}^{\text{Jdg}}$ ), Single Agent leads slightly (61.0% vs. 59.0% for CTU-Chat, 52.0% for Generic;  $\Delta = -2.0$  pp, 95% CI  $[-12.0, +8.0]$  pp). Single Agent also achieves higher Mean Answer Relevancy (80.9% vs. 77.8% for CTU-Chat, 66.7% for Generic). This stems from response characteristics: Single Agent generates concise, direct answers focused narrowly on the prompt, whereas CTU-Chat operates as an advisory counseling system incorporating regulatory constraints, prerequisite paths, and fee conditions, which incurs minor verbosity penalties from the judge despite preserving factuality. Under strict semantic criteria ( $\text{Pass}_{\geq .75}^{\text{Jdg}}$ ), CTU-Chat attains a higher point estimate (37.0% vs. 33.0% for Single Agent,  $\Delta = +4.0$  pp; 29.0% for Generic). However, a paired McNemar test reveals this difference is not statistically significant ( $p=0.4807$ , paired 95% bootstrap CI  $[-4.0, +12.0]$  pp spanning zero), indicating that strict semantic pass rates reflect an estimated point difference rather than confirmed semantic superiority. On financial queries ( $N_{\text{fin}}=50$ ), programmatic regex extraction and VND normalization confirm that deterministic calculation yields 62.5% Numeric Exact Match compared to 60.0% for Single Agent ( $\Delta = +2.5$  pp, 95% CI  $[-6.7, +11.8]$  pp) and 54.0% for Generic. These results demonstrate that specialist tools reduce arithmetic hallucination while general semantic performance exhibits nuanced trade-offs across answer conciseness and multi-agent coordination.

**Operational Trade-offs:** Table 1 delineates clear architectural trade-offs: Single Agent provides lower end-to-end latency (5,159 ms vs. 10,652 ms) and higher Answer Relevancy (80.9% vs. 77.8%) with zero supervisor overhead. Conversely, CTU-Chat achieves higher lexical fact pass (+10.0 pp,  $p=0.0414$ ), stronger cross-domain synthesis (80.0% vs. 70.0%), and reduces specialist input tokens by 41.4% (3,690 vs. 6,296, plus ~650 routing tokens).

## 5.2 Scenario 2 — Architectural Ablation

Table 2 removes one mechanism at a time from Full CTU-Chat ( $N=100$ ). Deltas are relative to the full system.

The largest pass-rate decreases occur without Specialist Policy ( $-10.0$  pp, 95% CI  $[-19.0, -1.0]$  pp) and with Uniform Text-RAG ( $-9.0$  pp, CI  $[-15.0, -4.0]$  pp). The former lowers Fact Coverage by 6.85 pp (CI  $[-13.0, -0.7]$  pp); the latter lowers Source Recall by 12.3 pp (CI  $[-16.5, -8.3]$  pp). Removing tool isolation leaves the pass estimate unchanged (CI  $[-6.0, +6.0]$  pp) but adds 1,270 input tokens. Without route repair, overall pass changes by  $-2.0$  pp (CI  $[-7.0, +3.0]$  pp), while cross-domain pass falls from 80.0% to 65.0%. Without deterministic calculation, overall automated fact pass changes by  $-4.0$  pp (95% CI  $[-9.0, 0.0]$  pp). This benchmark-wide result uses the  $\text{FactCov} \geq 0.50$  threshold and does not measure numeric exact match. Panel B shows 63.2% pass

**Table 2.** Scenario 2: Architectural ablation results ( $N = 100$ ). Panel A reports overall benchmark differences ( $\Delta$  relative to Full CTU-Chat). Panel B reports Automated Fact Pass Rate ( $\text{Pass}_{\geq 50\%}$ , %) across query families.

| <b>Panel A: Overall Benchmark Ablations</b> |                           |                     |                      |                    |                    |
|---------------------------------------------|---------------------------|---------------------|----------------------|--------------------|--------------------|
| <b>Ablated Component</b>                    | $\Delta\text{Pass}_{.50}$ | $\Delta\text{Fact}$ | $\Delta\text{Src.R}$ | $\Delta\text{Tok}$ | $\Delta\text{Lat}$ |
| No Spec. Policy                             | -10.0                     | -6.85               | 0.0                  | -217               | -1731              |
| Full Tool Vis.                              | 0.0                       | +0.93               | 0.0                  | +1270              | +296               |
| Uniform Text-RAG                            | -9.0                      | -2.18               | -12.3                | -300               | -1910              |
| No Route Rep.                               | -2.0                      | -1.27               | 0.0                  | -219               | -1610              |
| No Determ. Calc. <sup>†</sup>               | -4.0                      | -0.93               | 0.0                  | -782               | -4540              |

| <b>Panel B: Fact-Grounded Pass Rate (<math>\text{Pass}_{\geq 50\%}</math>) by Query Family</b> |                                  |                                       |                                     |                                  |
|------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------|-------------------------------------|----------------------------------|
| <b>Configuration</b>                                                                           | <b>Graph (<math>n=19</math>)</b> | <b>Struct-fin (<math>n=28</math>)</b> | <b>Narr-reg (<math>n=33</math>)</b> | <b>Cross (<math>n=20</math>)</b> |
| <b>Full CTU-Chat</b>                                                                           | 63.2                             | <b>75.0</b>                           | 81.8                                | <b>80.0</b>                      |
| No Spec. Policy                                                                                | 52.6                             | 67.9                                  | 72.7                                | 65.0                             |
| Full Tool Vis.                                                                                 | <b>68.4</b>                      | 67.9                                  | <b>84.8</b>                         | <b>80.0</b>                      |
| Uniform Text-RAG                                                                               | 63.2                             | 60.7                                  | 78.8                                | 60.0                             |
| No Route Rep.                                                                                  | <b>68.4</b>                      | <b>78.6</b>                           | 78.8                                | 65.0                             |

<sup>†</sup>Full benchmark with calculators disabled.  $\Delta\text{Tok}/\text{Lat}$  in tokens/ms per query.

on graph queries versus 80.0% on cross-domain queries. Graph queries can require credit aggregation across degree blocks; omitting one required fact may lower this substring-based score.

### 5.3 Scenario 3 — Tool Ambiguity Stress Test

Table 3 reports the semantic-neighbor experiment across 11–51 tools ( $N=50$ , 3 counter-balanced tool-order blocks, 1,800 primary decisions).

As the registry scales from 11 to 51 tools, Full-Registry Single Agent E2E success drops from 86.0% to 83.3% (input tokens +124%). Global Top-10 drops from 83.3% to 78.7% E2E (Gold Recall: 100.0% to 96.0%). Routed Specialist maintains 88.0% E2E at 51 tools (9.8 visible tools), yielding a +9.3 pp margin over Top-10 (95% CI [+2.0, +17.3] pp) and +4.7 pp over Full-Registry (95% CI [-1.3, +11.3] pp) with 52.4% fewer input tokens (1,770 vs. 3,722). Degradation relative to Top-10 is +7.3 pp (95% CI [0.0, +15.3] pp).

Across 1,800 primary decisions, Routed Specialist latency ranges from 1,983 to 2,021 ms (supervisor routing: ~950 ms; input tokens: 1,441–1,830). Full-Registry latency rises from 1,098 to 1,289 ms as input tokens scale from 1,663 to 3,722.

### 5.4 Supporting Retrieval Validation

Table 4 presents a standalone offline evaluation of retrieval ( $N=100$ ), isolating evidence retrieval performance from downstream multi-agent generation and routing. E4 cross-encoder reranks Hybrid RRF candidates; E5 reranks governed, hybrid, and BM25 candidates, then merges graph evidence under quotas.

**Table 3.** Scenario 3: Semantic-neighbor tool ambiguity ( $N = 50$ , 3 counterbalanced tool-order blocks).

| Reg. | Config.             | Vis. | Gold-R       | Sel.        | E2E         | Tok         | Lat         |
|------|---------------------|------|--------------|-------------|-------------|-------------|-------------|
| 11   | Full-reg single     | 11   | –            | <b>94.0</b> | <b>86.0</b> | 1663        | <b>1098</b> |
|      | Top-10 single       | 10   | <b>100.0</b> | 93.3        | 83.3        | 1541        | 1148        |
|      | Routed gen.         | 5.1  | 98.0         | 90.7        | 80.0        | <b>1204</b> | 1889        |
|      | <b>Routed spec.</b> | 5.1  | 98.0         | 93.3        | 85.3        | 1441        | 1983        |
| 31   | Full-reg single     | 31   | –            | 91.3        | 84.7        | 2706        | 1157        |
|      | Top-10 single       | 10   | <b>98.0</b>  | 90.0        | 81.3        | <b>1149</b> | <b>1033</b> |
|      | Routed gen.         | 9.8  | <b>98.0</b>  | 88.0        | 79.3        | 1396        | 1958        |
|      | <b>Routed spec.</b> | 9.8  | <b>98.0</b>  | <b>93.3</b> | <b>88.7</b> | 1830        | 2015        |
| 51   | Full-reg single     | 51   | –            | 92.7        | 83.3        | 3722        | 1289        |
|      | Top-10 single       | 10   | 96.0         | 86.7        | 78.7        | <b>1036</b> | <b>997</b>  |
|      | Routed gen.         | 9.8  | <b>98.0</b>  | 91.3        | 82.7        | 1340        | 1907        |
|      | <b>Routed spec.</b> | 9.8  | <b>98.0</b>  | <b>92.7</b> | <b>88.0</b> | 1770        | 2021        |

**Table 4.** Standalone retrieval evaluation ( $N=100$ ) on unique source filenames. E4: Hybrid RRF with cross-encoder reranking. E5: domain-governed, hybrid, and BM25 candidates with reranking and graph quotas. Evaluated independently of agent orchestration.

| Configuration             | Hit@1        | Hit@3        | P@5          | R@5          | MRR@10       |
|---------------------------|--------------|--------------|--------------|--------------|--------------|
| E1: BM25 Lexical          | 0.530        | 0.770        | 0.206        | 0.617        | 0.669        |
| E2: Dense Semantic        | 0.460        | 0.650        | 0.176        | 0.524        | 0.560        |
| E3: Hybrid RRF            | 0.520        | 0.790        | 0.224        | 0.662        | 0.673        |
| E4: Hybrid + Reranker     | 0.700        | 0.910        | 0.254        | 0.751        | 0.805        |
| <b>E5: Full Retrieval</b> | <b>0.740</b> | <b>0.920</b> | <b>0.302</b> | <b>0.861</b> | <b>0.833</b> |

BM25 outperforms dense search on Hit@1 (0.530 vs. 0.460), while Hybrid RRF balances both (0.790 Hit@3). The cross-encoder reranker yields the largest individual gain (+18.0 pp Hit@1). The full configuration reaches 0.740 Hit@1, 0.861 Recall@5, and 0.833 MRR@10; because it incorporates domain-specific components, the gain cannot be attributed to one component alone.

## 5.5 Failure Analysis

Table 5 categorizes errors in 58 failed query–configuration cases with Fact Coverage below 50% (34 for Single Agent and 24 for CTU-Chat). Labels were assigned through manual review of execution traces and intermediate outputs; categories are non-mutually exclusive.

The 34 Single Agent and 24 CTU-Chat failed query cases include 12 versus 3 tool errors. Retrieval errors are similar under shared evidence: 3 absent gold sources in each arm and 7 versus 6 reranker demotions. With complete retrieval ( $sr = 1.0$ ), partial fact omissions number 22 versus 14. These counts do not isolate effects of tool scoping or specialist prompts.

**Table 5.** Failure taxonomy across architecture configurations ( $N = 100$ ). Categories are non-mutually exclusive; a single query may exhibit multiple error modes.

| Category                                     | Failure Mode          | S1-A | S1-C | Category                 | Failure Mode          | S1-A                  | S1-C |
|----------------------------------------------|-----------------------|------|------|--------------------------|-----------------------|-----------------------|------|
| Routing                                      | Corrected misroute    | N/A  | 1    | Tool                     | Invalid arguments     | 4                     | 1    |
|                                              | Repair misroute       | N/A  | 1    |                          | Wrong-family call     | 3                     | 1    |
| Retrieval                                    | Gold source absent    | 3    | 3    | Generation               | Unnecessary call      | 5                     | 1    |
|                                              | Cross-domain omission | 3    | 2    |                          | Result ignored        | 2                     | 1    |
|                                              | Reranker demoted      | 7    | 6    |                          | Partial fact omission | 22                    | 14   |
| <b>Total Failed Queries (&lt; 50% Fact):</b> |                       |      |      | <b>S1-A (Single): 34</b> |                       | <b>S1-C (CTU): 24</b> |      |

Three traces illustrate distinct failure stages: in HOUT-MHOP-ACAD-04, repair mis-routed a `general` query to `academic`; in HOUT-XDOM-01, all gold sources were retrieved ( $sr=1.0$ ) but the worker omitted the financial answer; and in HOUT-XDOM-16, retrieval missed the target document ( $sr=0.5$ ).

## 6 Discussion

### 6.1 Architectural Findings and Operational Trade-offs

**Architectural Findings (RQ1–RQ3):** CTU-Chat achieves higher lexical pass (+10.0 pp) and uses 41.4% fewer specialist input tokens than Single Agent. Ablations confirm that specialist policies and heterogeneous evidence drive the largest gains. Ablating deterministic calculators lowers fact pass by an estimated  $-4.0$  pp (95% CI  $[-9.0, 0.0]$  pp); without route repair, cross-domain pass drops from 80.0% to 65.0%. At 11 tools, removing isolation increases input tokens without altering pass rates. At 51 tools, Routed Specialist achieves 88.0% tool E2E Success (+9.3 pp over Top-10, CI  $[+2.0, +17.3]$  pp; +4.7 pp over Single Agent, CI  $[-1.3, +11.3]$  pp), demonstrating robustness under ambiguity.

**Deployment Guidance and Trade-offs:** (1) *Single Agent* is preferred for homogeneous text policies without multi-hop prerequisites: bypassing supervisor routing reduces latency ( $\sim 5.2$  s vs. 10.65 s), saves  $\sim 650$  routing tokens, and achieves higher Answer Relevancy (80.9% vs. 77.8%) via concise answers without extraneous counseling context. It remains dependable for small tool pools ( $\leq 11$  tools, 94.0% selection). (2) *CTU-Chat* is recommended when queries involve heterogeneous representations (graphs, vector search, calculators) or cross-domain rules (80.0% vs. 70.0% pass). For high-stakes financial queries, it mitigates arithmetic hallucination, yielding higher strict semantic pass ( $\text{Pass}_{\geq 75\%}^{\text{Jdg}}$ : 37.0% vs. 33.0%, paired McNemar  $p=0.4807$ , 95% CI  $[-4.0, +12.0]$  pp) and numeric match (Num-EM: 62.5% vs. 60.0%, 95% CI  $[-6.7, +11.8]$  pp), though both intervals span zero. Under registry scaling ( $\geq 50$  tools), specialist scoping prevents selection collapse (88.0% vs. 78.7% E2E). Institutional transfer requires schema mapping, rule encoding, and routing calibration.

*Relation to Prior Work.* While ReAct [18], Toolformer [14], and AutoGen [16] examine generic coordination, CTU-Chat routes queries to bounded specialists with modality-

tailored evidence paths. Unlike Adaptive-RAG’s complexity retrieval [4], our approach isolates specialist registries, aligning with GraphRAG on relational evidence [2].

*Threats to Validity.* *Internal/operational:* 10.65 s latency limits interactivity; differing boundaries prevent direct speed rankings. *External:* Single-institution evaluation; transfer remains untested. Multi-turn history was omitted. *Construct:* Lexical matching can pass invalid outputs (e.g., HOUT-DIR-FIN-03 scored 50% despite a 192.183M vs. 40M VND error); programmatic evaluation penalized this (AC = 0.10, Num-EM = 0.0), catching arithmetic errors that substring overlap forgives (Num-EM: 62.5% vs. 60.0%). *Statistical:* Paired bootstrap analysis ( $N=100$ , 10,000 resamples); subsets have smaller samples ( $n = 19-33$ ).

## 7 Conclusion and Future Work

We evaluated CTU-Chat, a supervisor-routed multi-agent RAG system for university counseling, across architecture comparisons, ablations, and tool scaling. In architecture comparisons, CTU-Chat surpasses the Single Agent in lexical fact pass (+10.0 pp,  $p=0.0414$ ) with 41.4% fewer specialist input tokens, while exhibiting comparable semantic accuracy (57.7% vs. 58.6% Mean AC; 37.0% vs. 33.0% strict pass,  $p=0.4807$ ) and higher financial numeric exact match (62.5% vs. 60.0%). Ablations confirm that deterministic calculators and curriculum graphs mitigate arithmetic and prerequisite errors. At 51 tools, specialist isolation maintains 88.0% tool E2E Success. Future work will address dispatch latency (10.65 s) and institutional transfer.

**Disclosure of Interests.** The authors have no competing interests to declare that are relevant to the content of this article.

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